HAWK Fakultät Ingenieurwissenschaften und Gesundheit Göttingen

POLITECNICO MILANO 1863

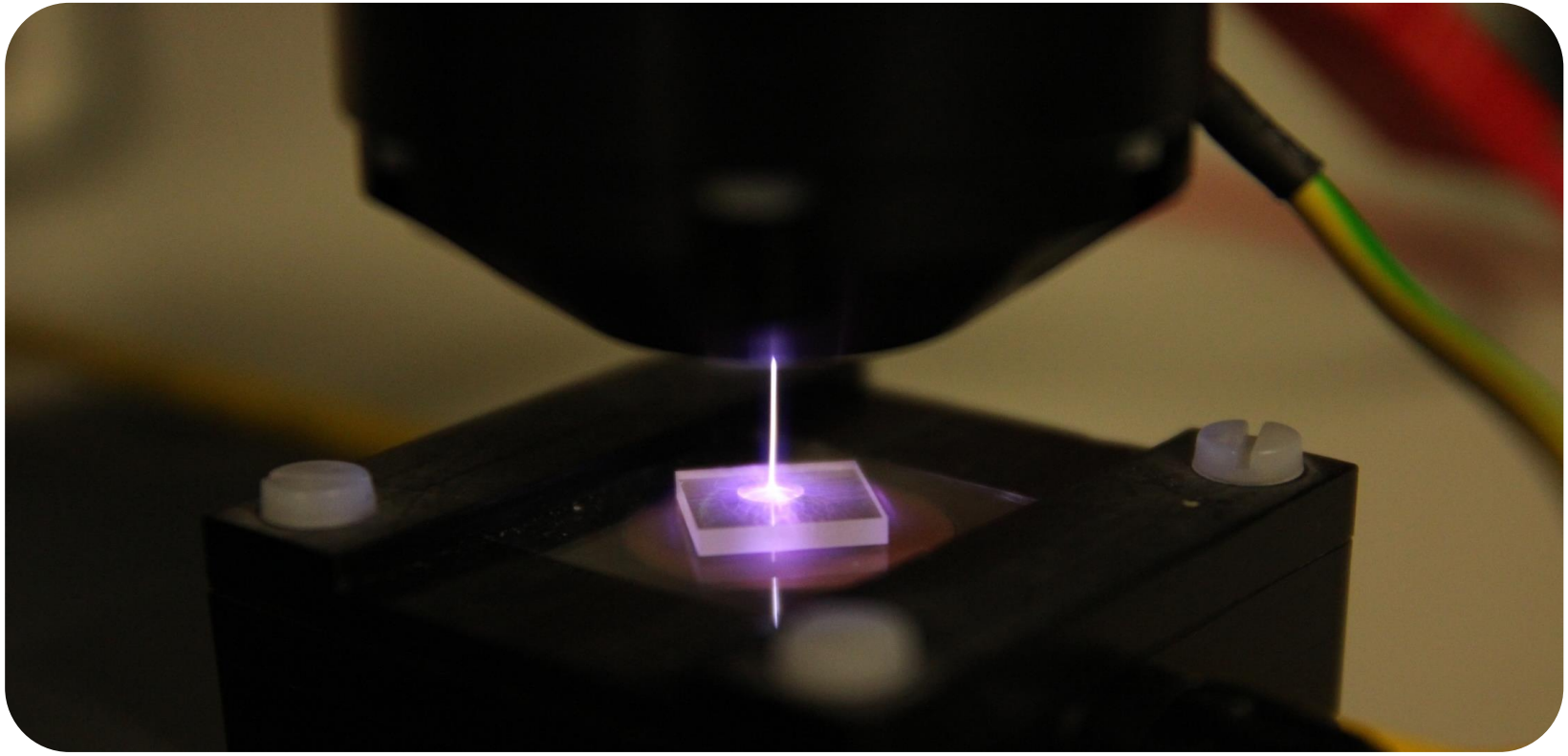
Exercises and Worked Examples + Solutions

ADVANCED TOPICS IN ENGINEERING PHYSICS

'Plasma and its applications'
Chapter 2

Prof. Dr. Christoph Gerhard, 2024

2 Plasma discharges and sources



Isothermal and equilibrium plasmas

???

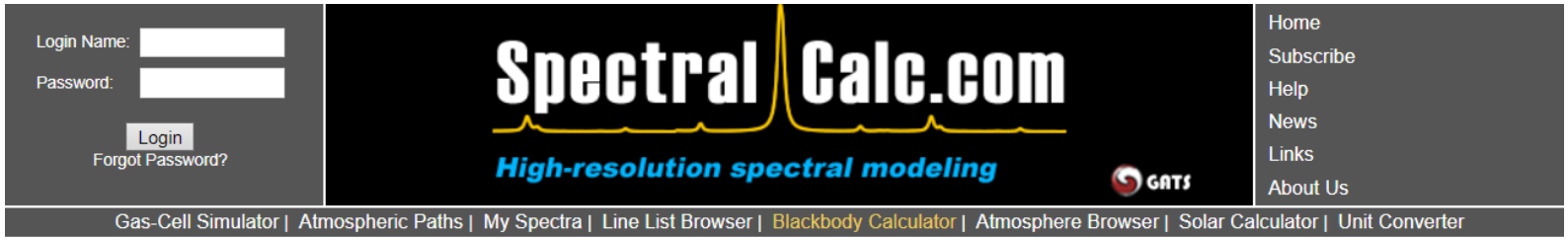
- **Exercise 2.1**
  - The spectral radiance of a blackbody at a temperature of 16000K amounts to $1.88102 \cdot 10^9$ $W \cdot m^{-2} \cdot sr^{-1} \cdot \mu m^{-1}$ at a wavelength of 350 nm. Determine the spectral radiance of the plasma at this wavelength assuming…
    a) …a uniform plasma in LTE with a temperature of T = 16000K, an absorption coefficient of α = 0.005/cm and a length of L = 12 mm and
    b) …a non uniform plasma consisting of two zones, i.e., a core (index c) and a peripheral zone (index p) where
       - T_c = 16000K, α_c = 0.008/cm and L_c = 6 mm and
       - T_p = 3000K, α_p = 0.002/cm and L_p = 6 mm and

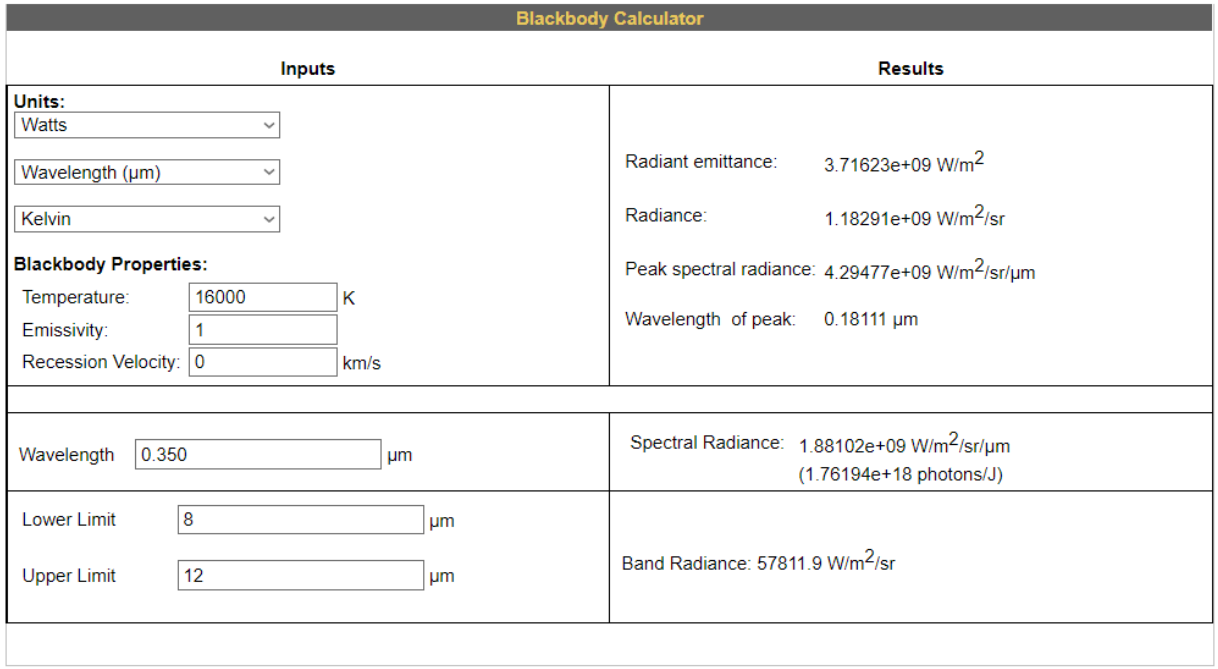
  - Practice tip: Blackbody Calculator
    https://www.spectralcalc.com/blackbody_calculator/blackbody.php

Isothermal and equilibrium plasmas

! ! !

- **Exercise 2.1 – Appendix**
  - Practice tip: Blackbody Calculator (https://www.spectralcalc.com/blackbody_calculator/blackbody.php)





Isothermal and equilibrium plasmas

! ! !

- **Exercise 2.1 – Solution**

a) Uniform plasma (no units displayed for calculation):

$$B = U \cdot (1 - e^{-\alpha \cdot L}) = 1.88102 \cdot 10^9 \cdot (1 - e^{-0.005 \cdot 1.2}) = 1.12523 \cdot 10^7$$

→ The spectral radiance is $1.12523 \cdot 10^7$ W·m^{-2}·sr^{-1}·µm^{-1}.

b) Non-uniform plasma (no units displayed for calculation):

- Here, U_p, where T_p amounts to 3000K, is 25389 W·m^{-2}·sr^{-1}·µm^{-1}.

$$B = U_c \cdot (1 - e^{-\alpha_c \cdot L_c}) \cdot e^{-\alpha_p \cdot L_p} + U_c \cdot \left(1 - e^{-\alpha_p \cdot L_p}\right)$$

$$B = 1.88102 \cdot 109 \cdot (1 - e^{-0.008 \cdot 0{,}6}) \cdot e^{-0.002 \cdot 0.6} + 25389 \cdot (1 - e^{-0.002 \cdot 0.66})$$

$$B = 9.00726 \cdot 10^6 \cdot 0.99880 + 30.4485 = 8.99648 \cdot 10^6$$

→ The spectral radiance is $8.99648 \cdot 10^6$ W·m^{-2}·sr^{-1}·µm^{-1}.

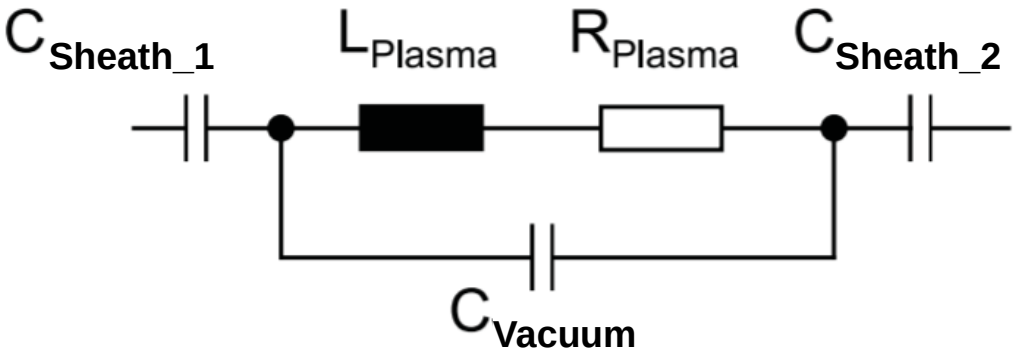
Conclusion: The difference in spectral radiance amounts to about $2.25582 \cdot 10^6$ W·m^{-2}·sr^{-1}·µm^{-1}.

→ Notable impact on evaluation of laser-induced plasmas! (see 4.2)

DC discharges

???

- **Exercise 2.2**
  - As a simple approach, a plasma source can be described by an equivalent circuit diagram as shown below, schematising a capacitive plasma generator.



  - We now assume that a plasma source is merely characterised by an electrical resistance of R = 1.7 MΩ. It is moreover operated at an applied voltage of U = 170 V.
    a) Which type of plasma is ignited in this case?
    b) Which type of plasma is ignited if the same voltage is applied, but the resistance is merely R = 90 Ω?

DC discharges

! ! !

- **Exercise 2.2 – Solution**

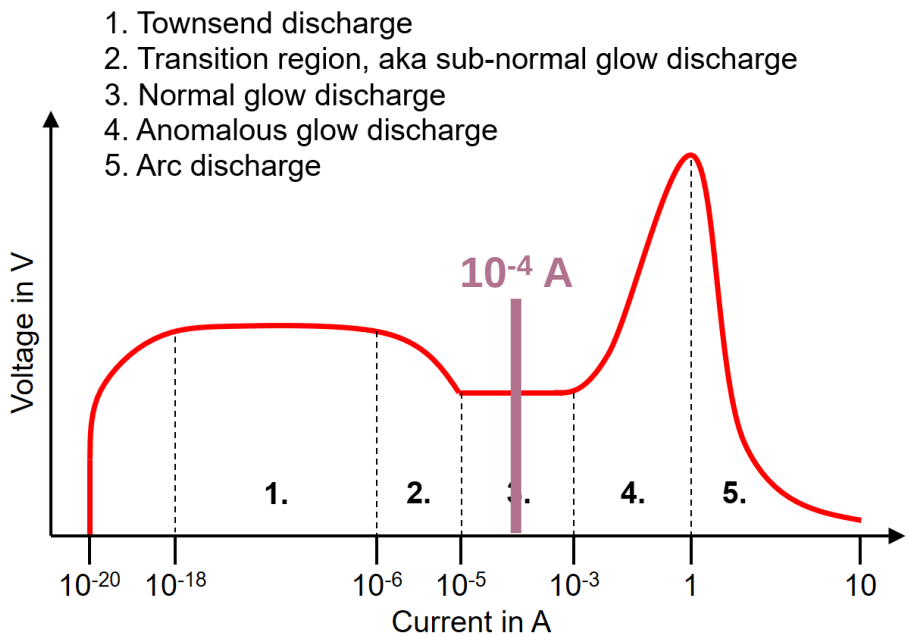
a) Applying Ohm's law,

$$U = R \cdot I,$$

The discharge current can be calculated by inserting the given values. It then amounts to

$$I = \frac{U}{R} = \frac{170V}{1.700.000\Omega} = 0{,}0001A = 10^{-4}A.$$

Thus, a normal glow discharge is formed:



DC discharges

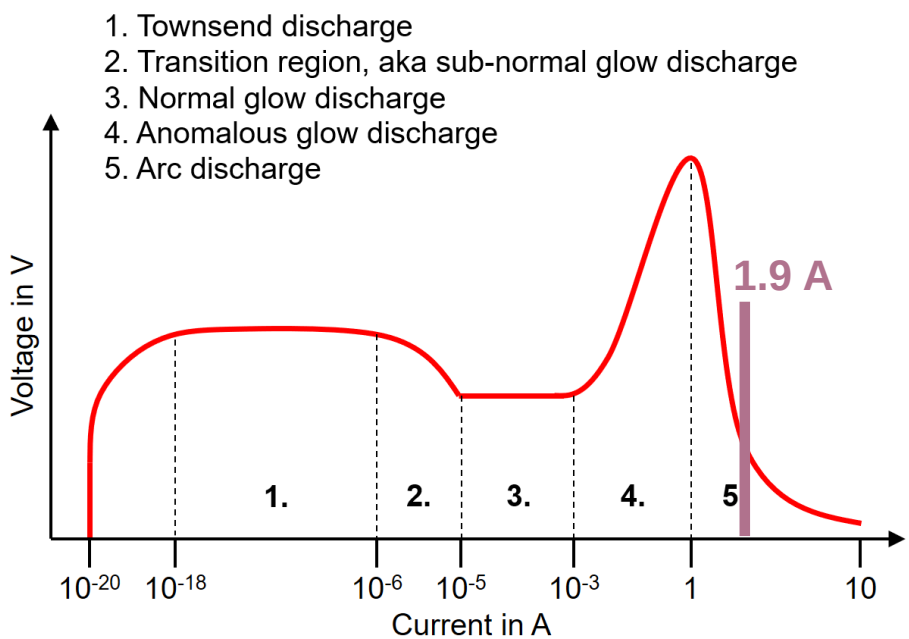
! ! !

- **Exercise 2.2 – Solution**

b) In this case, we obtain

$$I = \frac{U}{R} = \frac{170V}{90\Omega} = 1.9A.$$

Hence, the plasma is an arc discharge:



The Townsend discharge

???

- **Exercise 2.3**
  - Molecular nitrogen is exposed to cosmic irradiation with a wavelength of 62 nm. Is molecular nitrogen ionised or not by the incident cosmic irradiation?

  - Practice tip: NIST Chemistry WebBook: https://webbook.nist.gov/cgi/cbook.cgi?ID=C7727379&Mask=20

The Townsend discharge

! ! !

- **Exercise 2.3 – Solution**
  - First, the ionisation energy of molecular nitrogen needs known.
  - Hint: NIST databases, here NIST Chemistry WebBook: https://webbook.nist.gov/cgi/cbook.cgi?ID=C7727379&Mask=20

NIST National Institute of Standards and Technology U.S. Department of Commerce — NIST Chemistry WebBook, SRD 69

Search ▾ | NIST Data ▾ | About ▾

Nitrogen

- **Formula:** N_2
- **Molecular weight:** 28.0134
- **IUPAC Standard InChI:** InChI=1S/N2/c1-2 InChI v 1.06
- **IUPAC Standard InChIKey:** IJGRMHOSHXDMSA-UHFFFAOYSA-N
- **CAS Registry Number:** 7727-37-9
- **Chemical structure:** N≡N

  This structure is also available as a 2d Mol file or as a computed 3d SD file

  The 3d structure may be viewed using Java or Javascript.
- **Other names:** Nitrogen gas; N2; UN 1066; UN 1977; Dinitrogen; Molecular nitrogen; Diatomic nitrogen; Nitrogen-14
- **Permanent link** for this species. Use this link for bookmarking this species for future reference.
- **Information on this page:**
  - Gas phase ion energetics data
  - References
  - Notes

The Townsend discharge

! ! !

- **Exercise 2.3 – Solution**
  - Ionisation energy of molecular nitrogen from NIST Chemistry WebBook:
  - 15.7 eV (highest value reported in literature = 'worst case')

Ionization energy determinations

IE (eV)	Method	Reference	Comment
15.581 ± 0.008	S	Trickl, Cromwell, et al., 1989	*LL*
15.7 ± 0.1	EI	Stephen, Mark, et al., 1984	*LBLHLM*
15.6 ± 0.1	EI	Grade, Wienecke, et al., 1983	*LBLHLM*
15.60	PE	Kimura, Katsumata, et al., 1981	*LLK*
10.1 ± 0.6	EI	Armentrout, Tarr, et al., 1981	*LLK*
15.58	EI	Armentrout, Tarr, et al., 1981	*LLK*
15.5808	EVAL	Huber and Herzberg, 1979	*LLK*
15.58 ± 0.02	EI	Sahini, Constantin, et al., 1978	*LLK*
15.5	PI	Rabalais, Debies, et al., 1974	*LLK*
15.58	PE	Lee and Rabalais, 1974	*LLK*
15.61	PE	Natalis, 1973	*LLK*
15.58 ± 0.01	PE	Hotop and Niehaus, 1970	*RDSH*
15.56	CI	Cermak, 1968	*RDSH*
15.58	PI	Cook and Metzger, 1964	*RDSH*
15.5803	S	Ogawa and Tanaka, 1962	*RDSH*
15.5802	S	Worley, 1943	*RDSH*
15.5812 ± 0.0002	S	Worley and Jenkins, 1938	*RDSH*
15.58	PE	Potts and Williams, 1974	Vertical value; *LLK*
15.60	PE	Katrib, Debies, et al., 1973	Vertical value; *LLK*

The Townsend discharge

! ! !

- **Exercise 2.3 – Solution**
  - Calculation of energy provided by incident cosmic irradiation via

$$E = h \cdot f = \frac{h \cdot c}{\lambda}$$

h = Planck's constant ≈ 6.63 x 10^{-34} Js
c = speed of light ≈ 3 x 10^{8} m/s

  - Hence,

$$E = h \cdot f = \frac{h \cdot c}{\lambda} \approx 3.2 \cdot 10^{-18} J \approx 20 eV.$$

1 eV ≈ 1.6 x 10^{-19} J

  - Molecular nitrogen is thus most likely ionised by the incident cosmic irradiation, since E_{ion} = 15.7 eV < E = 20 eV.

The Townsend discharge

???

- **Exercise 2.4**
  - Determine the energy of an electron, which is generated via photoelectric emission at a copper surface where the wavelength of incident light is 210 nm.

The Townsend discharge

! ! !

- **Exercise 2.4 – Solution**
  - The energy provided by the incident light (= photon energy) amounts to

$$E_{ph} = h \cdot f = \frac{h \cdot c}{\lambda} \approx 9.5 \cdot 10^{-19} J \approx 5.9 eV.$$

h = Planck's constant ≈ 6.63 x 10^{-34} Js
c = speed of light ≈ 3 x 10^{8} m/s
1 eV ≈ 1.6 x 10^{-19} J

  - Since the work function of copper ranges from 4.53 to 5.10 eV, the electron energy (or kinetic energy of the electron) ranges from

    <u>0.8 eV</u> to **<u>1.37 eV</u>**

    according to

$$E_{e-} = E_{kin} = E_{ph} - W.$$

The Townsend discharge

???

- **Exercise 2.5**
  - Determine the wavelength of incident light required to remove an electron from an aluminium surface via photoelectric emission where the resulting energy of the removed electron should be sufficient for ionising atomic hydrogen.

The Townsend discharge

! ! !

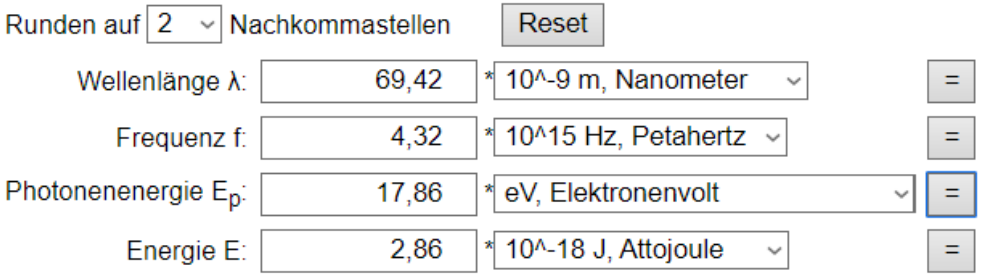
- **Exercise 2.5 – Solution**
  - The energy for ionising hydrogen – and thus the minimum energy of the electron – amounts to 13.6 eV. The work function of aluminium varies from 4.06 eV to 4.26 eV…so we choose the higher value as the worst case scenario. The required photon energy follows from

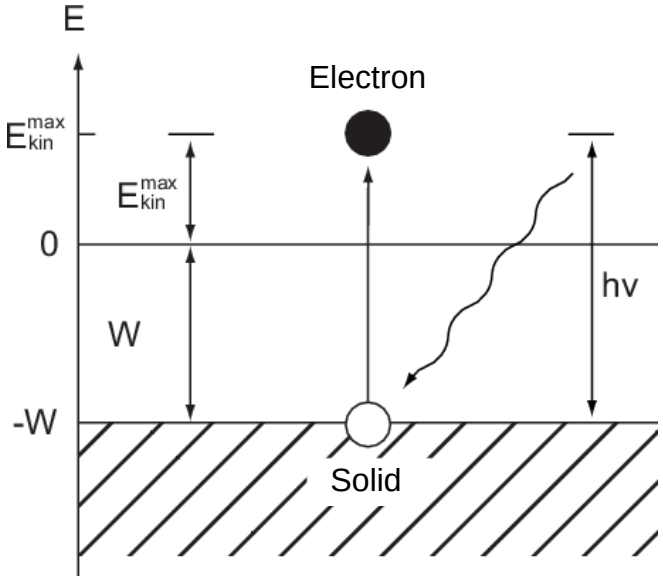
$$E_{e-} = E_{kin} = E_{ph} - W$$

  - and is

$$E_{ph} = W + E_{e-} = 17.86eV.$$

  - This value corresponds to a wavelength of **<u>69.42 nm</u>** ≈ 70 nm.
  - Helpful online tool (in German): https://rechneronline.de/spektrum/





The Townsend discharge

???

- **Exercise 2.6**
  - In a plasma source, the discharge gap, i.e., the distance between the electrodes, amounts to 1 cm. We assume that an initial electron, which is emitted by the cathode, causes ionisation, and that each electron resulting from such ionisation features sufficient energy for ionisation of the gas (here: a mixture of 80 % nitrogen and 20% oxygen) between the electrodes. The plasma is operated at
    a) atmospheric pressure,
    b) a pressure of 1 hPa, and
    c) a pressure of 10^{-7} hPa.

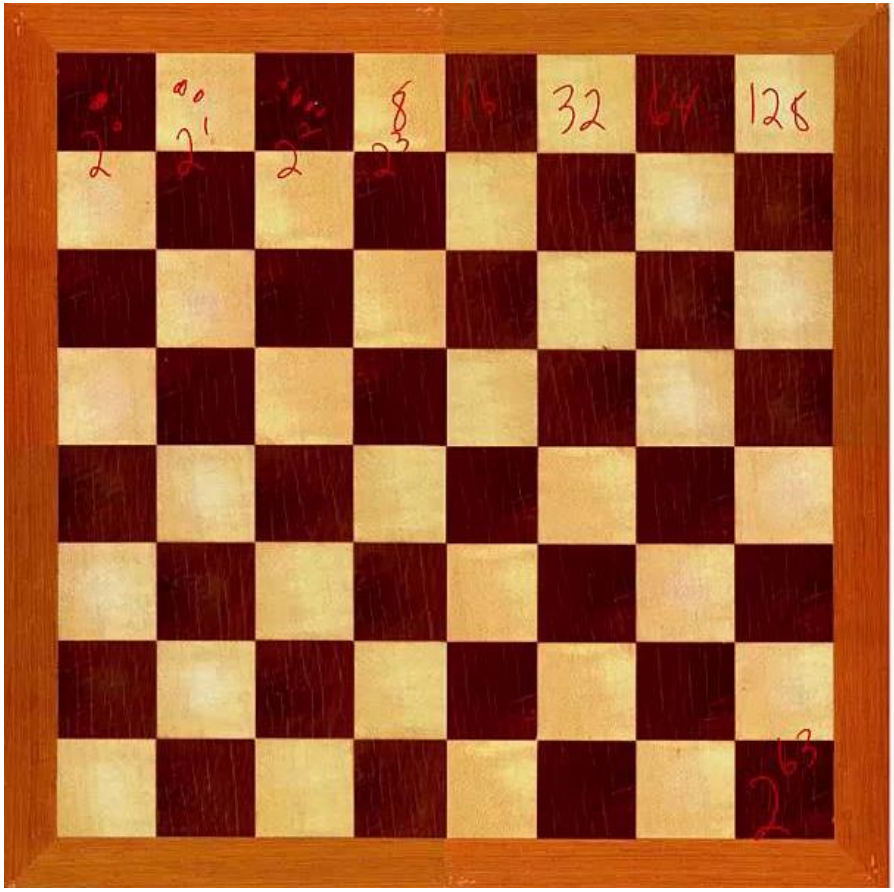
  - Estimate the minimum number of free electrons generated between both electrodes in case a), b), and c).

The Townsend discharge

! ! !

- **Exercise 2.6 – Solution**

Mathematical approach: ‘Wheat and chessboard problem’



How many grains of wheat would be on the chessboard at the end?

$1+2+2^2+2^3+\ldots+2^{63} = 2^{64}-1$:

18,446,744,073,709,551,615

That's more than 18 quintillions!

CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=48783038

The Townsend discharge

! ! !

- **Exercise 2.6 – Solution**

'Minimum number of free electrons' → electrons generated along the direct connection (= straight line) between involved electrodes, i.e., the discharge gap thickness

a) Atmospheric pressure = 1013 hPa → λ = 68 nm (see 1.2.5)
→ number of collisions N = 1 cm/68 nm = 10000000/68 = 147059
→ $N_e = 2^{147059}-1 \approx \infty$ (infinite)

b) 1 hPa → λ = 100 µm (see 1.2.5)
→ number of collisions N = 1 cm/100 µm = 10000/100 = 100
→ $N_e = 2^{100}-1 \approx 1.3^{30}$

c) 10^{-7} hPa → λ = 1 km (see 1.2.5)
→ number of collisions N = 1 cm/1 km = 0.01/1000 = 0.00001
→ $N_e = 2^{0.00001}-1 \approx 0$

(However, the initial free electron is still present…so the '-1' does not apply)

The Townsend discharge

???

- **Exercise 2.7**
  - A plasma is ignited in pure molecular hydrogen as process gas. The discharge gap width is 2.54 cm and the applied voltage amounts to 12 kV. Moreover, the plasma is operated at atmospheric pressure. Is this configuration suitable for an ionisation of the process gas?

The Townsend discharge

! ! !

- **Exercise 2.7 – Solution**

For solving this task, the electron energy has to be determined. For this purpose, the 1rst Townsend coefficient is calculated initially. It amounts to

$$\alpha = A \cdot p \cdot e^{-\left(\frac{B \cdot p \cdot d}{V}\right)} = \frac{3.83}{m \cdot Pa} \cdot 10^5 Pa \cdot e^{-\left(\frac{109.1 \frac{V}{Pa \cdot m} \cdot 10^5 Pa \cdot 0.0254 m}{12 \cdot 10^3 V}\right)} = 1.03 \cdot 10^{-7} m^{-1}$$

Since the 1rst Townsend coefficient is also given by

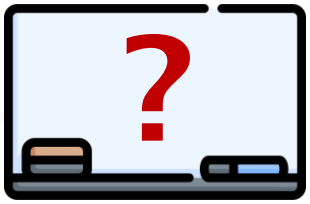
$$\alpha = \frac{1}{\lambda} \cdot e^{\left(\frac{E_{ionisation}}{E_{e^-}}\right)}$$

with λ being the mean free length of path at atmospheric pressure (68 nm, see 1.2.5), the energy of an electron amounts to E_{e^-} = 0.6 eV:

$$\alpha = \frac{1}{\lambda} \cdot e^{\left(\frac{E_{ionisation}}{E_{e^-}}\right)} \rightarrow \frac{E_{ionisation}}{E_{e-}} = 25.68 \rightarrow E_{e-} = \frac{E_{ionisation}}{25.68} = \frac{15.4 eV}{25.68} \approx 0.6 eV$$

This energy is 25.68-times lower and thus not sufficient for ionisation of molecular hydrogen since the required energy is 15.4 eV.

Paschen's law



- **Worked example 2.1**
  - In the following table, different values for the ignition voltage and $p \cdot d_{min}$ are displayed for the gas mercury…why?

Gas	**Cathode material**	**Ignition voltage @ $p \cdot d_{min}$ in V**	**$p \cdot d_{min}$ in Pa·m**
Argon	Iron	265	2.0
Helium	Iron	150	3.3
Mercury	Iron	**520**	**2.6**
Mercury	Tungsten	**425**	**2.36**
Nitrogen	Iron	275	1.0
Oxygen	Iron	450	0.92

Paschen's law

- **Worked example 2.1 – Answer**
  - Even though the values refer to the same gas, the electrode material differs (iron vs. tungsten). Since the Paschen's minimum, given by

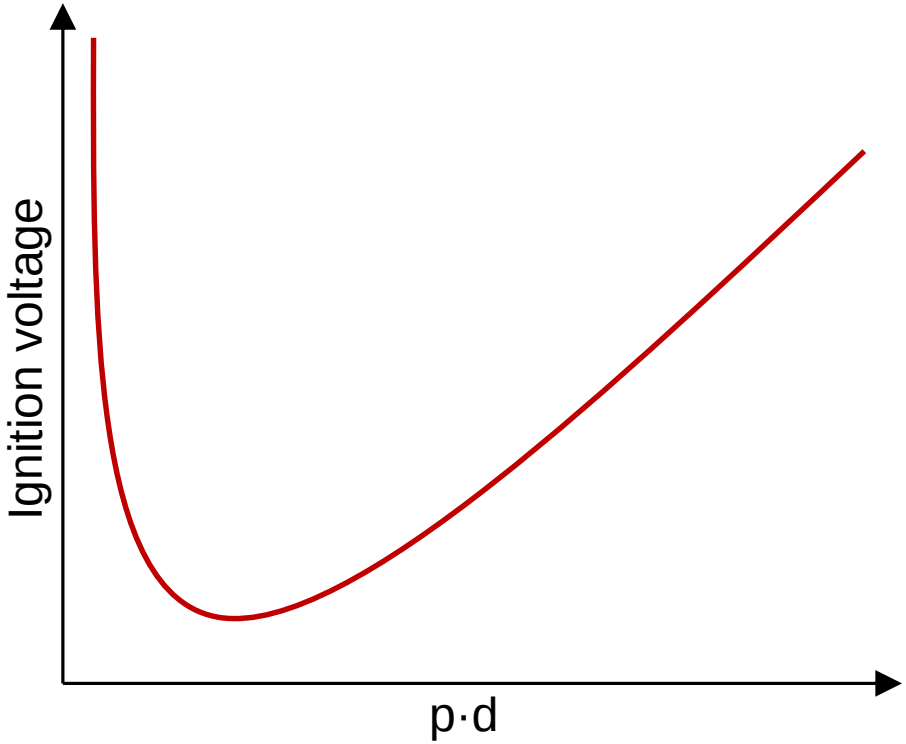
    $$Apd - \ln(1+\gamma^{-1}) = 0$$

    depends on the second Townsend coefficient and thus the electrode material different values result consequently.

Gas	**Cathode material**	**Ignition voltage @ $p \cdot d_{min}$ in V**	**$p \cdot d_{min}$ in Pa·m**
Argon	Iron	265	2.0
Helium	Iron	150	3.3
Mercury	**Iron**	520	2.6
Mercury	**Tungsten**	425	2.36
Nitrogen	Iron	275	1.0
Oxygen	Iron	450	0.92

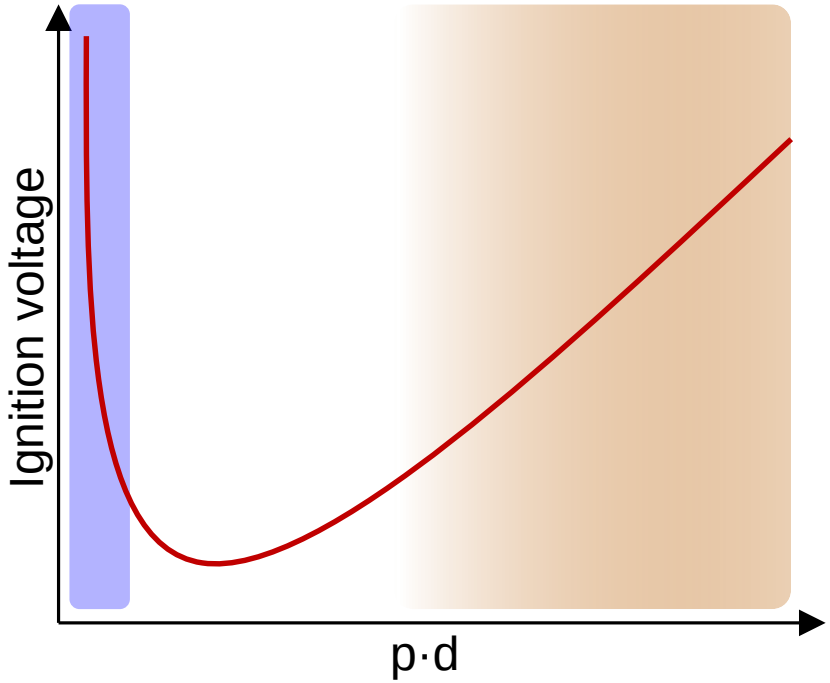
Paschen's law

- **Worked example 2.2**
  - As expressed by the Paschen's curve, a very high ignition voltage is required for very low and very high p·d-values (and discharge gap thicknesses in case of constant pressure, respectively)…why?



Paschen's law

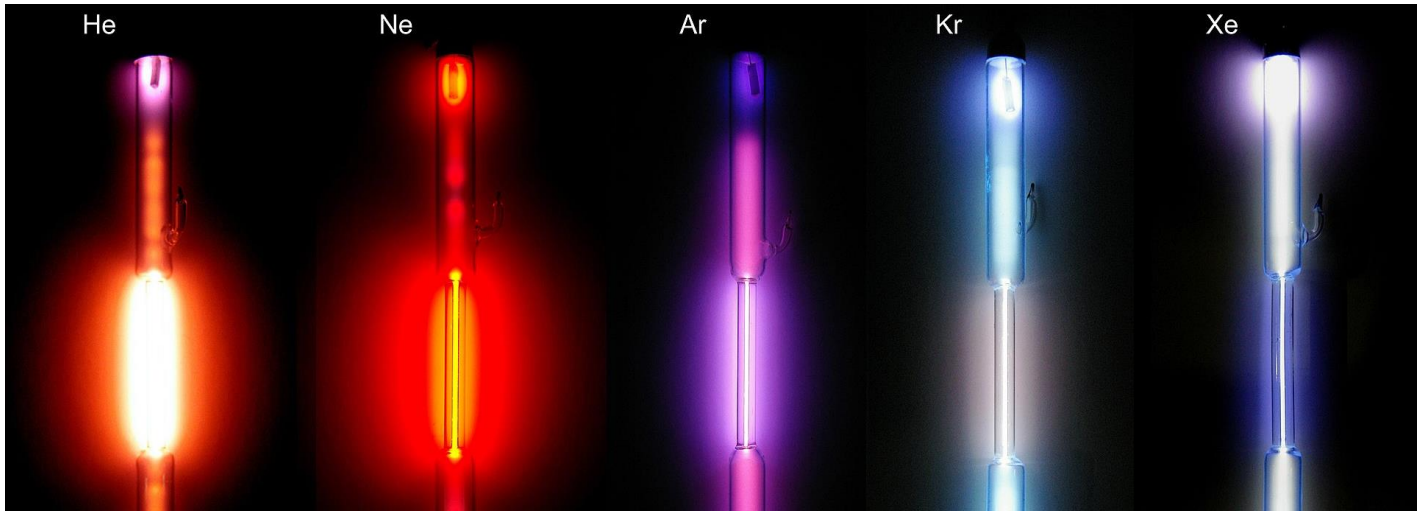
- **Worked example 2.2 – Answer**
  - Very low p·d-values:
    At such short distances, electrons cannot gain sufficient energy for ionisation and generation of further electrons, respectively
  - Very high p·d-values:
    The higher the value, the longer the time of flight of electrons and thus the number of collisions and loss in electron energy, respectively



Paschen's law

- **Worked example 2.3**
  - A light source, based on the principle of fluorescent tubes, shall be realised considering the following requirements:
    - Gas pressure within the tube: 1000 Pa
    - Colour of light emitted by gas discharge: bright red-orange
    - Cathode material: iron
    - Required voltage: as low as possible

a) Which gas could be used to fill the tube?
b) Determine the optimum length of the tube.

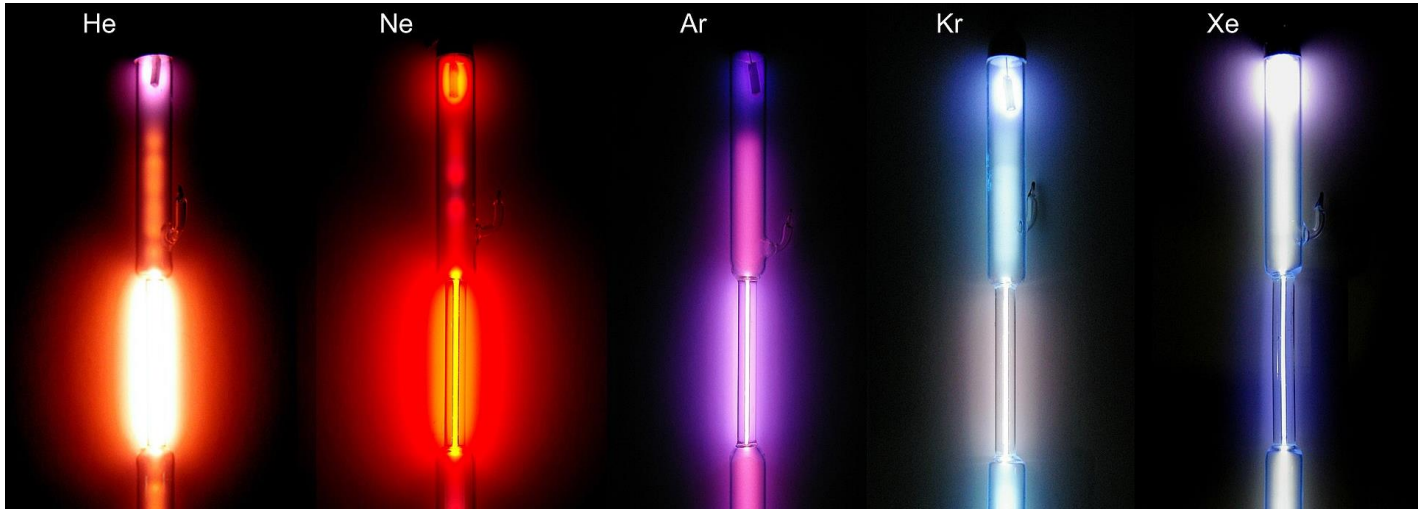


Paschen's law

- **Worked example 2.3 – Answer**

a)
Colour of light emitted by gas discharge: bright red-orange

→ Helium (or Neon)



Paschen's law

- **Worked example 2.3 – Answer**

b)

- Colour of light emitted by gas discharge: bright red-orange
- Cathode material: iron
- Required voltage: as low as possible

Gas	**Cathode material**	**Ignition voltage @ $p \cdot d_{min}$ in V**	**$p \cdot d_{min}$ in Pa·m**
Argon	Iron	265	2.0
Helium	Iron	150	3.3
Mercury	Iron	520	2.6
Mercury	Tungsten	425	2.36
Nitrogen	Iron	275	1.0
Oxygen	Iron	450	0.92

Paschen's law

- **Worked example 2.3 – Answer**

b)

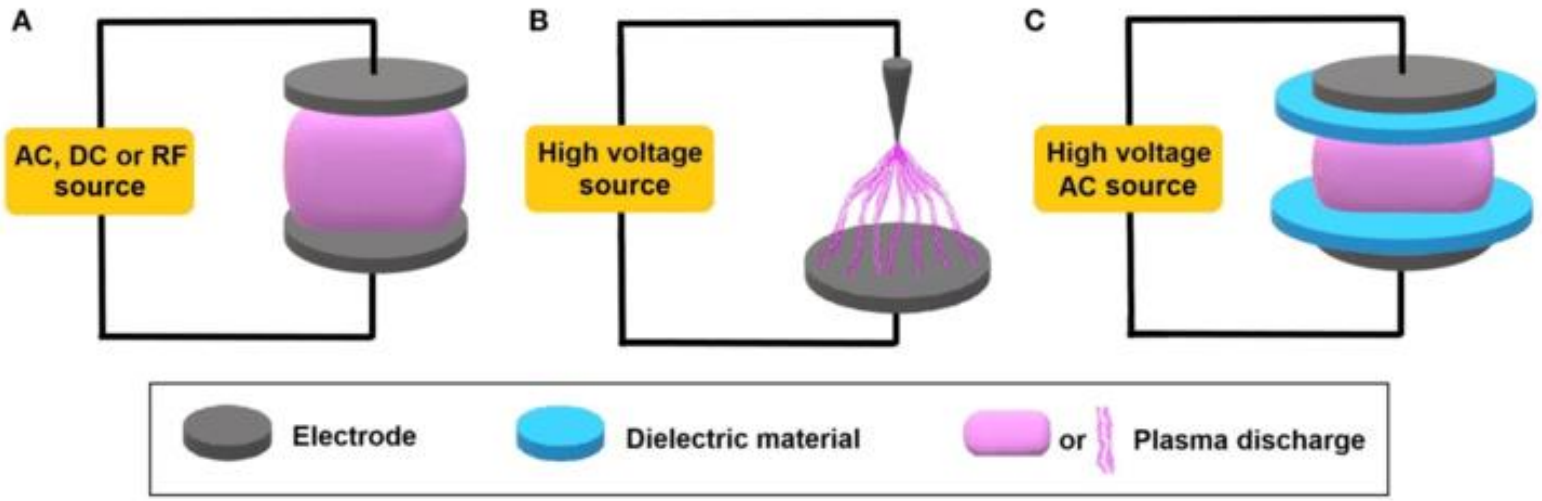
- Gas pressure within the tube: 1000 Pa

Gas	**Cathode material**	**Ignition voltage @ $p \cdot d_{min}$ in V**	**$p \cdot d_{min}$ in Pa·m**
Argon	Iron	265	2.0
Helium	Iron	150	3.3
Mercury	Iron	520	2.6
Mercury	Tungsten	425	2.36
Nitrogen	Iron	275	1.0
Oxygen	Iron	450	0.92

→ Optimum length of the tube corresponds to discharge gap thickness at Paschen's minimum d_{min} = 3.3 Pa·m/1000 Pa = <u>3.3 mm</u>.

Dielectric barrier discharges

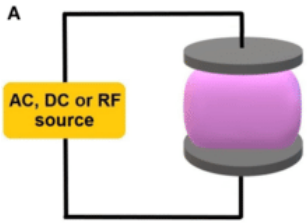
- **Worked example 2.4**
  - Suggest some technical applications for the three different types of plasma shown below.



DOI:10.3389/fmech.2020.00042

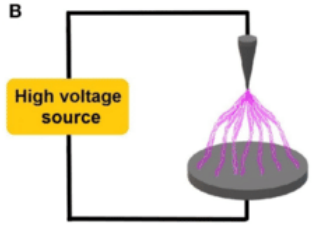
Dielectric barrier discharges

- **Worked example 2.4 – Answer**
  - A = hot and homogeneous discharge



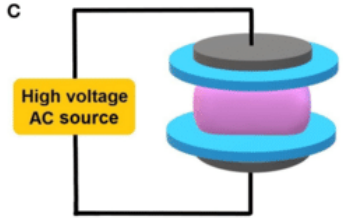
    - Applications: welding and cutting
    - (classical circuit of plasma flame cutter)
    - Device: ignition plug

  - B = hot, filamentary, and inhomogeneous (plasma streamers with high energy density)



    - Cell perforation
    - Surface structuring
    - etc.

  - C = cold and homogenous



    - Gentle surface activation and functionalisation (of temperature-sensitive materials, e.g. plastics or papers)
    - Plasma medicine
    - Surface cleaning
    - etc.

Dielectric barrier discharges

???

- **Exercise 2.8**
  - Calculate the optimum discharge gap thickness of a dielectric barrier discharge plasma at atmospheric pressure where the discharge gap is filled with helium and the electrodes are covered with sodium glass plates as dielectric. For solving this task, the following table was found in literature:

Table 1 Paschen coefficients for several commonly used gases (taken from [4]).

gas	electrode	A $[Pa^{-1}m^{-1}]$	B $[VPa^{-1}m^{-1}]$	γ
He	sodium - glass	2.10	25.50	0.263
N_2 (air)	platinum	10.95	273.78	0.025
Ar	tungsten	10.20	176.27	0.095

[4] J. Cobine, ”Gaseous conductors”, (Dover, New York, 1958).

Dielectric barrier discharges

! ! !

- **Exercise 2.8 – Solution**

The optimum discharge gap thickness is found at the Paschen's minimum, given by

$$A \cdot p \cdot d - \ln(1 + \gamma^{-1}) = 0. \quad (1)$$

The pressure p is constant at atmospheric pressure and thus 1013 hPa. The gas-dependent coefficient *A* and the 2nd Townsend coefficient *γ* are taken from literature where $A = 2.1\ (\mathrm{Pa \cdot m})^{-1}$ [a] (or $A = 2.25\ (\mathrm{Pa \cdot m})^{-1}$ [b]) and $\gamma = 0.263$, respectively. Solving Eq. (1) for the wanted discharge gap thickness *d* and inserting these values gives

$$d = \frac{\ln(1 + \gamma^{-1})}{A \cdot p} = \frac{\ln(1 + 0.263^{-1})}{2.1(Pa \cdot m)^{-1} \cdot 1.013^{5} Pa} = 7.4 \cdot 10^{-6} m = 7.4 \mu m.$$

The optimum discharge gap thickness, where the lowest ignition voltage is required, is thus 6.8 µm in the given case. But in practice, this value is the cathode fall and APP discharge gaps are thicker (some hundreds of microns to some cm).

Note: For $A = 2.25\ (\mathrm{Pa \cdot m})^{-1}$, it amounts to 6.8 µm.

[a]J. Cobine, "Gaseous conductors", (Dover, New York, 1958)
[b] J. Lehr, P. Ron: Electrical breakdown in Gases. In: Foundations of pulsed power technology. John Wiley $ Sons, Inc., 2017

Dielectric barrier discharges

???

- **Exercise 2.9**
  - An atmospheric pressure plasma in local thermal equilibrium should be used for surface activation of a plane work piece made of PVC, where the operating temperature of PVC is 60°C (above this temperature, the work piece made of PVC deforms). The plasma is operated at a high voltage of 15 kV; the discharge gap thickness is 3 cm.

  - Can such a plasma be used for the described application?

Dielectric barrier discharges

! ! !

- **Exercise 2.9 – Solution**

The essential required information is the plasma temperature. For its determination, the electron energy is calculated as follows:

$$E = U/d = 15\ \text{kV}/3\ \text{cm} \approx 500000\ \text{V/m}.$$

and

$$E_{e^-} = \lambda \cdot E = 68 \cdot 10^9\ \text{m} \cdot 500000\ \text{V/m} \approx 0.034\ \text{V} = 0.034\ \text{eV}.$$

The temperature of the LTE plasma is thus 11604K (= 1 eV)·0.034 = 394K, corresponding to about 121°C…which is twice the limit of 60°C.

→ The LTE plasma cannot be used for the described activation of PVC!

But what type of plasma could be used instead?